AAQ Physics
Pilot implementation brief

20-mark sessions • Lesson-based topic choices • Transparent method credit

15 October 2026 | Teacher review and development handoff

For the Pearson AAQ Applied Science Extended Certificate, starting with Waves. Revision for a cohort aiming for Merit, with accessible selected answers and scaffolded calculations.

Agreed behaviour

• Use a 20-mark session target, with a variable number of whole questions. Marks make workload more comparable; they do not guarantee equal completion time.

• Aim for 50% recall by question count, not by marks. The revised sample bank has 28 questions: 14 recall and 14 application.

• Use exam-like command words in questions. Use concise, natural English in feedback.

• Automate longer answers using selected missing sentences. Include exactly one open-response self-assessment test.

• Keep the established chemistry tool design, including its newer dark mobile bar, green controls and pink/red Finish control.

Session construction

1. All required lessons selected; enabled questions only; correct review mode.

2. Whole questions, no duplicate IDs or two variants from the same family.

3. Reach exactly 20 marks when possible; otherwise use the largest attainable total below 20 and announce it before starting. Never split a question to fill the budget.

4. Among sets at that total, minimise the absolute difference between recall fraction and 0.5.

5. Prefer less-seen questions as a tie-break, then seeded random order.

A short-pool message must appear before starting: “Only 12 marks are available from these lessons. Start this shorter session or select more lessons.” Do not include untaught content or repeat a question to make up the total.

Student journey

Choose studied lessons → see the actual mark total → start practice → check each answer → read correction and explanation → review the session and revisit mistakes.

Pack status

Content, lesson map, schema and tested reference selection/marking logic are supplied. Browser interface implementation and classroom testing follow this brief. All questions remain drafts for teacher review.

Waves: retain your 12 headings

Most headings are one lesson-sized choice. Total Internal Reflection and Uses of Waves open to reveal three choices each. Harmonics and communication may each need two teaching slots. These are planning estimates, not Pearson allocations.

ID / selectable lesson	Scope	Specification
WA01 Types of Waves	Transverse/longitudinal waves; energy transfer; compressions and rarefactions.	A1.2
WA02 Wave Properties	Amplitude, wavelength, frequency, period, oscillation and speed; definitions and units.	A1.1.1
WA03 Graphing Waves	Read and distinguish displacement-distance and displacement-time graphs.	A1.1.2
WA04 Wave Speed	$v = f\lambda$; $f = 1/T$; rearrangement, units and conversions.	A1.4
WA05 Phase Difference	Displacement, coherence, phase/path difference; constructive/destructive superposition.	A1.3
WA06 Diffraction	Double-slit interference application; transfer to grating contexts; link to line spectra.	A1.3, A1.3.2
WA07 Emission Spectra	Electron transitions, photons, grating separation and identification from spectral lines.	A1.3.1, A1.3.2, A1.3.3
WA08 Standing Waves	Incident/reflected waves; resonance; displacement nodes and antinodes.	A1.5
WA09 Harmonics	String/pipe harmonics and pitch; $v = \sqrt{(T/\mu)}$; mass per unit length.	A1.5.1, A1.5.2
WA10 Refraction	TIR group: refraction, ray direction, $n = c/v$ and air-to-material sine ratios.	A2.1, A2.1.1
WA11 Critical Angle and TIR	TIR group: both conditions, critical angle and glass-air calculations.	A2.1.2, A2.1.3
WA12 Optical Fibres and Cladding	TIR group: core/cladding boundaries, repeated TIR and fibre applications.	A2.1.4, A2.2
WA13 Inverse Square Law	Relative intensity; $I = k/r^2$; rearrangement, units and ideal-model assumptions.	A3.2
WA14 Electromagnetic Spectrum	Uses group: EM region order, overlap, frequency/wavelength and vacuum speed.	A3.1, A3.3
WA15 Analogue and Digital Signals	Uses group: analogue/digital representations, binary waveforms and regeneration.	A2.3
WA16 Communication Applications	Uses group: satellite/GPS, mobile phones, Bluetooth, infrared and Wi-Fi.	A3.3.1

Approximately 18 slots of 50-60 minutes. The Diffraction heading retains the requested double-slit interference application; grating context and line spectra remain in the scope notes. No compulsory white-light rainbow task or fringe-spacing equation is added.

Forces: proposed lesson choices

These sections follow the specification and are reserved for later question-bank expansion. They should not start empty sessions in the Waves pilot.

ID / lesson	Scope	Specification
FO01 Motion Quantities and Units	SI symbols/units; speed conversions; scalar/vector distinctions.	B1.1, B1.3.1, B1.3.2
FO02 Speed and Average Speed	Speed = distance/time; multi-stage average speed.	B1.2
FO03 Displacement-Time Graphs	Interpret displacement-time traces; signed velocity from gradients.	B1.3.3
FO04 Velocity-Time Graphs and Acceleration	Interpret velocity-time traces; acceleration from gradient and a = (v-u)/t.	B1.3.4, B1.3.6
FO05 Distance and Displacement from Graphs	Areas under velocity-time graphs; signed displacement versus total distance.	B1.3.5
FO06 Motion Equations: Velocity and Average Velocity	Constant acceleration; v = u + at and s = (u+v)t/2.	B1.3.8
FO07 Motion Equations: Displacement	Constant acceleration; $s = ut + \frac{1}{2}at^2$ and $v^2 = u^2 + 2as$; signs and units.	B1.3.8
FO08 Measuring Acceleration and Accelerometers	Trolley investigation; measurements and evidence; accelerometer applications.	B1.3.7, B1.4
FO09 Resultant Forces and Newton’s First Law	Force diagrams; balance and constant velocity; resultant force changes velocity.	B2.1, B2.7
FO10 Mass, Weight and Inertia	Mass versus weight; inertia; gravitational field strength and W = mg.	B2.2
FO11 Friction and Its Measurement	F = μN; normal reaction; static/kinetic friction measurement.	B2.3, B2.3.1, B2.3.2
FO12 Newton’s Second Law	F = ma with resultant force; effects of changing mass and acceleration.	B2.5, B2.5.1
FO13 Momentum	p = mv; units, direction and mass/speed comparisons.	B2.4
FO14 Force and Change in Momentum	F = (mv-mu)/t; direction, stopping time and average impact force.	B2.5
FO15 Safety and Impact Protection	Transport comparisons; air bags, seat belts, helmets, passenger cells and crumple zones.	B2.5.2, B2.5.3
FO16 Newton’s Third Law	Interaction pairs on different bodies; distinguish pairs from balanced forces.	B2.6
FO17 Drag and Terminal Velocity	Drag and terminal velocity: road vehicles, parachutes and objects falling in liquids.	B2.7.1

Approximately 19 slots of 50-60 minutes. Allow extra review and assessment time; practical sections may need longer. Newton’s first and second laws use resultant force; third-law pairs act on different bodies.

Electricity: proposed lesson choices

These sections follow the specification and are reserved for later question-bank expansion. They should not start empty sessions in the Waves pilot.

ID / lesson	Scope	Specification
EL01 Circuit Symbols, Quantities and Units	Symbols and definitions; current, p.d., energy, power, resistance and units.	C1.1, C1.2, C1.2.1
EL02 Connecting Circuits and Meters	Build circuits; ammeters in series; voltmeters in parallel.	C1.3, C1.3.1
EL03 Resistance and V = IR	Ohmmeter use on isolated, unpowered components; V = IR.	C1.3.2, C2.1.2
EL04 Filament Lamps and Diodes	Lamps and diodes in circuits; interpret supplied evidence.	C1.4
EL05 Thermistors and LDRs	Temperature/light-dependent resistance and sensor applications.	C1.4
EL06 Photodiodes and LEDs	Photodiodes detect light; LEDs emit light; use in circuits.	C1.4
EL07 Electrical Power	P = IV and P = E/t; rearrangement and units.	C2.1.1, C2.1.3
EL08 Electrical Energy	E = VIt; time conversions and calculations.	C2.1.4
EL09 Domestic Energy Use	Appliance comparisons; kWh = kW × h; use supplied tariffs if needed.	C3.1, C3.3
EL10 Fuses and Operating Current	Operating current and supplied fuse ratings; justify selection.	C3.2
EL11 Energy Units and Temperature Scales	J, kJ, MJ; Celsius/Kelvin conversion and temperature changes.	C4.1, C4.2
EL12 Temperature Change and Change of State	Heating within a phase versus melting/boiling; heating/cooling evidence.	C4.3
EL13 Specific Heat Capacity Calculations	$\Delta Q = mc\Delta T$; rearrangement; J kg^{-1} K^{-1}.	C4.4.2, C4.4.3
EL14 Measuring Specific Heat Capacity	Solid/liquid methods; electrical heating, measurements, losses and evaluation.	C4.4.1
EL15 Specific Latent Heat Calculations	$\Delta Q = mL$; fusion/vaporisation; mass and J kg^{-1}.	C5.2
EL16 Measuring Specific Latent Heat	Fusion/vaporisation methods; measurements, losses and evaluation.	C5.1

Approximately 18 slots of 50-60 minutes. Allow extra review and assessment time; practical sections may need longer. Electricity includes thermal energy, specific heat capacity and latent heat as required by section C.

Marking students can understand

Worked ECF example: W03

Correct calculation: 85 kHz = 85 000 Hz; λ = 340/85 000 = 0.0040 m = 4.0 mm.

Suppose a student enters 85 Hz, selects λ = v/f, then obtains 4 m and 4000 mm. The unit conversion at the start is wrong; their later division and conversion follow that value correctly.

Part	Student response	Credit
Frequency in Hz	85	0/1: original conversion error
Rearrangement	λ = v/f	1/1: direct credit
Wavelength in metres	4	1/1: method credit (ECF)
Wavelength in millimetres	4000	1/1: method credit (ECF)

Show: “3/4 marks: 1 direct mark + 2 method marks (error carried forward). kilo means 1000, so the frequency should be 85 000 Hz. You divided by your frequency correctly, then converted your metres to millimetres correctly.”

Marking rules

• Check the whole question once to avoid revealing answers to later parts. Warn about blanks and allow an intentional partial submission.

• Use only the authored ECF dependencies. Blank, zero-denominator, non-finite or invalid-domain values cannot seed credit. Award any mark only once.

• A correct later value can receive direct credit even if an earlier box is wrong. Independent calculations stay independent.

• Accept equivalent permitted units and numerical formats. Keep unrounded intermediate values when available.

• Keep the first attempt, hint use, reveals and retries distinct. Show direct and ECF marks separately; self-assessed marks remain a separate result.

Preserve the existing interface

Retain the v1.4.1 navy sidebar, light workspace, serif headings, green actions, existing question feedback and session navigation. Keep mobile Change Topics and Mistakes green, and Finish Session pink/red. Add the physics controls in the existing answer area. Keep instructions prominent, labels explicit and diagrams enlargeable.

Use AAQ-specific save keys so chemistry progress is unaffected. Ask before changing the navigation, palette, page hierarchy or phone interaction pattern. Do not add account or cloud features to this pilot.

AUTOMATED LONGER RESPONSE | TEACHER REVIEW

W12: finish the communication answer

W12 | 4 practice marks | Compare

A technician sends instructions to a display 4 m away. People regularly pass between the transmitter and receiver. The table shows successful transmissions out of 10. Compare the methods, explain the effect of the obstruction and suggest the more suitable method for this room. Complete the answer by selecting one sentence for each gap.

Condition	Infrared	Bluetooth
No obstruction	10/10	10/10
Plastic screen	2/10	9/10

Supplied opening - no marks

Both methods transmitted successfully in all 10 trials without an obstruction.

Gap 1: Compare the obstructed results.

A. With the screen, infrared succeeded in 2/10 trials and Bluetooth in 9/10 trials.
B. With the screen, infrared succeeded in 9/10 trials and Bluetooth in 2/10 trials.
C. With the screen, both methods still succeeded in all 10 of the trials.
D. With the screen, infrared succeeded in 8/10 trials and Bluetooth in 1/10 trials.

Gap 2: Explain the infrared result.

A. Infrared needs no clear path, so the screen should not affect reception.
B. Infrared needs a clear path, and the screen blocked much of its signal.
C. Infrared cannot travel 4 m, even when there is no screen in the path.
D. Infrared has a lower frequency after the screen, so the receiver stops working.

Gap 3: Explain the Bluetooth result.

A. Bluetooth uses infrared waves that are unaffected by this screen.
B. Bluetooth travels faster than infrared, reaching the receiver before an obstruction matters.
C. Bluetooth gains energy in the plastic screen, increasing the strength of the signal.
D. Bluetooth uses radio waves that passed through this screen well enough for reception.

Gap 4: Suggest a method and justify it for this room.

A. Use infrared because both methods were equally successful when the path was clear.
B. Use Bluetooth because its signal will pass through every material in the room.
C. Use Bluetooth because interruptions are likely and it worked better with an obstruction.
D. Use infrared because fewer successful transmissions mean less unwanted interference.

Answer key and marking

1 A; 2 B; 3 D; 4 C. One independent mark per selected missing point. The opening sentence is supplied and earns no marks. Options are shuffled in use; marking uses option IDs.

Distractor focus

The wrong options target swapped data, successes versus failures, clear-path results used in an obstructed setting, changed frequency, faster electromagnetic travel, energy gain and overgeneralising to every material. Each has its own corrective feedback in the structured bank.

AUTOMATED LONGER RESPONSE | TEACHER REVIEW

W27: finish the optical-fibre explanation

W27 | 3 practice marks | Explain

An optical fibre has core refractive index 1.50 and cladding refractive index 1.40. The critical angle at this boundary is 69°. A ray reaches the boundary from inside the core at an incidence angle of 75°. Explain how the light remains in the core. Complete the answer by selecting one sentence for each gap.

Supplied opening - no marks

The ray reaches the core-cladding boundary from inside the core.

Gap 1: Compare the refractive indices.

A. The core has a lower refractive index than the cladding.
B. The core and cladding have equal refractive indices.
C. The core has a greater refractive index than the cladding.
D. Light travels faster in the core than in the cladding.

Gap 2: Relate the incidence angle to the critical angle.

A. The incidence angle is greater than the critical angle at this boundary.
B. The incidence angle must equal the critical angle at this boundary.
C. The incidence angle is less than the critical angle at this boundary.
D. Any non-zero incidence angle is sufficient at this boundary.

Gap 3: Explain how the light remains in the core.

A. Light refracts into the cladding and then reflects from the outer surface.
B. Light is absorbed by the core and re-emitted at each boundary.
C. Light is totally internally reflected back into the core at successive boundaries.
D. Light is reflected because it approaches the boundary from the lower-index material.

Answer key and marking

1 C; 2 A; 3 C. One independent mark per missing point: direction of refractive index, incidence exceeding the critical angle, and repeated TIR at the core-cladding boundary. The supplied sentence earns no marks.

Why these distractors matter

Index comparison: reverses higher/lower index, treats unequal values as equal, or reverses the index-speed relationship.

Angle condition: confuses above the critical angle with equal to it, below it, or simply non-zero.

Mechanism: substitutes an outer-surface reflection, absorption/re-emission, or the wrong approach medium for TIR.

All options are complete sentences that fit the answer position. Each wrong sentence expresses a distinct error; feedback supplies the correct point before explaining it. These are guided practice marks, not evidence of independent extended-writing performance.

W25: one written self-assessment test

W25 | 4 practice marks | Compare

A museum guide sends information to a nearby display. Visitors frequently walk between the devices. The table shows successful transmissions out of 12. Compare the methods, explain the effect of the obstruction and suggest the more suitable method for the museum. Write your answer before revealing the marking points.

Condition	Infrared	Bluetooth
Clear path	12/12	12/12
Obstructed path	3/12	11/12

Student workflow

Write in a labelled text box → submit and preserve the first answer → reveal the bullet marking points → enter a whole-number score in “Your self-assessed score” (0 to 4) → continue. Place the score box at the end of this marking workflow. Keep the answer and score when resuming.

Bullet marking points

• 1 mark: Compare both results with the obstruction: infrared 3/12 and Bluetooth 11/12 (or equivalent percentages).

• 1 mark: Explain that the obstruction blocks much of the infrared signal / interrupts the clear line of sight.

• 1 mark: Explain that Bluetooth uses radio waves able to pass through this particular obstruction sufficiently well.

• 1 mark: Recommend Bluetooth and link the recommendation to visitors frequently passing between the devices; do not claim that it passes through every material.

Worked answer

Both methods worked in all 12 trials with a clear path. With an obstruction, infrared succeeded in 3/12 trials and Bluetooth in 11/12. The obstruction blocks much of the infrared signal, which needs a clear line of sight. Bluetooth uses radio waves that could pass through this obstruction well enough for reception. Bluetooth is the better choice here because visitors frequently interrupt the path and the obstructed results show greater reliability.

Result handling

Validate the score as an integer from 0 to 4. A blank score is not zero. The tool must not claim to have graded the prose. Record the result as self-assessed and report it separately from automatic marks. W12 and W25 are related variants and must not appear together in a session.

Accessibility

Use a persistent field label, a clear maximum and a specific validation message. Keep typed writing when validation fails. Provide keyboard access and readable focus states; the phone keyboard must not obscure the score box or Continue button.

Development handoff and question inventory

The bank contains 60 practice marks across 28 questions: 22 selected-answer/completion tasks, five scaffolded calculations and one written test. Default sessions select whole questions up to 20 marks.

ID	Question	R/A · marks	ID	Question	R/A · marks
W01	Read a wave graph	A · 1	W15	Recall the wave equation	R · 1
W02	Describe a longitudinal wave	R · 2	W16	Recall a transverse-wave definition	R · 1
W03	Calculate an ultrasound wavelength	A · 4	W17	Recall coherence	R · 1
W04	Explain double-slit interference	A · 3	W18	Recall the principle of superposition	R · 1
W05	Identify gases from emission spectra	A · 3	W19	Recall a displacement node	R · 1
W06	Read a standing-wave pattern in a pipe	A · 2	W20	Recall resonance	R · 1
W07	Calculate string tension and pitch	A · 4	W21	Recall the two conditions for TIR	R · 2
W08	Choose the refracted ray	A · 1	W22	Recall analogue and digital signals	R · 2
W09	Calculate a critical angle	A · 3	W23	Recall electromagnetic-wave speed	R · 1
W10	Recognise and regenerate a binary signal	A · 2	W24	Use two different wave graphs	A · 4
W11	Apply the inverse-square law	A · 3	W25	Compare methods: written self-assessment test	A · 4
W12	Compare communication methods using evidence	A · 4	W26	Order electromagnetic waves	R · 1
W13	Recall frequency and periodic time	R · 2	W27	Explain transmission in an optical fibre	A · 3
W14	Recall amplitude and wavelength	R · 2	W28	Recall refraction	R · 1

Validation and next build

The supplied reference module passes 19 test groups, including 100 seeded full-topic sessions, ECF, accepted numeric formats, dependent marks and self-score validation. Data structure and cross-references are checked separately. These checks do not replace browser QA.

Next: connect these render/marking contracts to the preserved UI; test phone layout, keyboard focus, diagram enlargement, partial submissions, save/resume and accessible feedback; then run teacher and small-cohort review. Questions remain drafts until reviewed.

Source and scope

Pearson Applied Science Extended Certificate specification, Issue 7 (May 2026): Unit 3 content, printed pp. 52-56; command terms, pp. 58-59. Lesson boundaries, practice marks and AO tags are authored judgements. The full source register is in the question-bank file. The supplied earlier Unit 3 extract was checked alongside the current specification.